

zkBugs Evaluation Results

Table 1: Tool Performance Summary

Tool	TP	FN	Timeout	Failure	Total	Median Time (s)
circumspect	0	70	0	0	70	0.03
circum_civer	12	11	22	25	70	0.88
picus	10	35	19	6	70	2.62
ecneproject	25	38	6	1	70	22.21
zkfuzz	25	16	25	4	70	0.97
conscs	11	49	10	0	70	2.88
TOTAL	83	219	82	36	420	1.98

Table 2: Execution Time Statistics

Tool	Total	Min (s)	Max (s)	Median (s)	Mean (s)	NR-Timeout
circumspect	69	0.01	0.33	0.03	0.07	0
circum_civer	23	0.04	502.33	0.88	86.00	22
picus	51	1.08	494.72	2.62	77.59	19
ecneproject	63	12.93	148.34	22.21	30.10	6
zkfuzz	41	0.01	531.69	0.97	40.41	25
conscs	60	0.23	544.52	2.88	59.30	10

Table 3: Bug-Tool Matrix (by Bug ID)

Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
1 FN	FN	FN	FN	FN	FN
2 Timeout	FN*	Timeout	FN	Timeout	Timeout
3 FN*	FN	Timeout	Timeout	TP*	TP*
4 FN	FN	FN	FN	FN	TP*
5 FN*	FN	FN	Failure	FN	TP*
6 FN	FN	Failure	FN	FN	Failure
7 TP*	FN	Failure	Timeout	TP*	Failure
8 FN	FN*	Failure	FN	FN	Failure
9 FN*	FN	TP	TP*	TP*	TP*
10 FN	FN*	Timeout	Timeout	TP*	TP*
11 TP*	FN	TP	TP*	TP*	TP*
12 FN	FN	Timeout	FN	FN	Timeout
13 FN	FN	Timeout	FN	FN	Timeout
14 Timeout	FN	FN	Timeout	FN	TP*
15 FN	FN	FN	Timeout	FN*	Timeout
16 FN	FN	FN	Timeout	FN	TP*
17 FN*	FN	Timeout	Timeout	TP*	FN
18 FN	FN	FN	FN*	FN*	Timeout
19 Timeout	FN	FN	FN	FN	Timeout
20	FN	Timeout	Timeout	Timeout	FN

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Bug ID	circumspect	circom_civer	picus	ecneproject	zkfuzz
FN*					
21	FN*	Timeout	FN	Timeout	Failure
FN					
22	FN*	Timeout	Timeout	Timeout	FN
FN*					
23	FN*	Timeout	Timeout	FN	Timeout
FN					
24	FN*	TP	TP*	TP*	TP*
TP*					
25	FN	Failure	FN	FN	FN
FN					
26	FN*	Failure	FN*	FN*	Timeout
FN					
27	FN*	Failure	FN	FN	FN
FN					
28	FN	Failure	FN	FN	Timeout
FN					
29	FN*	Failure	FN*	FN*	Timeout
FN					
30	FN	TP	TP*	TP*	TP*
TP*					
31	FN*	Failure	FN*	FN*	TP*
FN					
32	FN*	FN	FN*	FN*	FN*
FN					
33	FN	Failure	FN*	Timeout	FN
Timeout					
34	FN	Timeout	Timeout	TP*	Timeout
FN*					
35	FN	Timeout	FN	FN	Timeout
FN					
36	FN	FN	FN	FN	Timeout
Timeout					
37	FN	Failure	FN	Timeout	FN
Timeout					
38	FN	Failure	FN	FN	Timeout
FN					
39	FN	Timeout	FN*	FN*	Timeout
FN					

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
40 FN	FN*	FN	FN	FN	Timeout
41 FN*	FN*	Timeout	Timeout	TP*	TP*
42 FN	FN	Timeout	Timeout	FN	Timeout
43 FN*	FN	TP*	Failure	TP*	TP*
44 FN*	FN	TP*	Failure	TP*	TP*
45 FN*	FN	TP*	Failure	TP*	TP*
46 TP*	FN	TP	TP*	TP*	TP*
47 TP*	FN	TP	TP*	TP*	TP*
48 TP*	FN	TP	TP*	TP*	TP*
49 Timeout	FN	Timeout	TP*	TP*	TP*
50 FN	FN*	Failure	FN	FN	Timeout
51 FN	FN	Failure	FN	FN	FN
52 FN	FN	Failure	FN	FN	FN
53 TP*	FN*	Timeout	Timeout	TP*	FN
54 FN	FN	Failure	FN	FN	TP*
55 Timeout	FN	Failure	FN	Failure	FN
56 FN	FN	Timeout	FN	FN	Timeout
57 Timeout	FN	Timeout	Timeout	TP*	Timeout
58 FN*	FN	Timeout	Timeout	TP*	TP*
59	FN*	TP	TP*	TP*	TP*

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
TP*					
60	FN*	Timeout	Failure	TP*	TP*
TP*					
61	FN	Failure	Timeout	TP*	Timeout
FN*					
62	FN*	Failure	FN*	FN*	FN*
FN					
63	FN*	Failure	FN*	FN*	FN*
FN					
64	FN	Failure	FN	FN	FN
FN					
65	FN*	Failure	Failure	FN	Timeout
FN					
66	FN	Failure	Timeout	TP*	Timeout
FN*					
67	FN	Failure	FN	FN	Timeout
FN					
68	FN	Timeout	Timeout	FN	Timeout
Timeout					
69	FN*	Failure	FN	FN	TP*
FN					
70	FN*	TP	TP*	TP*	TP*
TP*					

* = *manually analyzed*

Table 4: Bug ID to Name Mapping

ID	Bug Name
1	daira_hopwood_darkforest_v0_3_missing_bit_length_check
2	hashcloak_data_are_not_fully_verified_during_state_update
3	kobi_gurkan_mimc_hash_assigned_but_not_constrained
4	koukyosyumei_checkdepth_comparator_overflow
5	koukyosyumei_clamp_comparator_overflow
6	koukyosyumei_emitifinrange_lessthan_overflow
7	koukyosyumei_extractstringfrompoint_shiftedfirstbyte_unconstrained
8	koukyosyumei_spend_missing_range_check
9	koukyosyumei_subtract_unconstrained_multiplier
10	koukyosyumei_unconstrained_base64_decoded_len
11	koukyosyumei_under_constrained_date_encoder
12	leastauthority_Circuit_Does_Not_Check_the_ERC_20_Sum_Correctly
13	leastauthority_previously_correct_ownership_proof_disabled_via_code_changes
14	matterlabs_email_spoofing_via_manipulated_from_header
15	matterlabs_emailauth_fails_with_overlapping_invitation_code_regex
16	matterlabs_fromaddrregex_allows_email_address_spoofing
17	matterlabs_underconstrained_fpmul_circuit
18	openzeppelin_mismatched_base64url_decoding_may_break_completeness
19	openzeppelin_non_determinism_of_some_inputs
20	trailofbits_incorrect_handling_of_point_doubling_can_allow_signature_forgery
21	trailofbits_prover_can_lock_user_funds_by_including_ill-formed_bigints_in_public_key_commitment
22	trailofbits_prover_can_lock_user_funds_by_supplying_non-reduced_Y_values_to_G1BigIntToSignFlag
23	trailofbits_unsafe_use_of_num2bits_in_multiple_circuits
24	veridise_arrayxor_is_under_constrained
25	veridise_babyjubjub_suborder_constraints_not_applied_correctly
26	veridise_blacklist_states_not_representable_in_field
27	veridise_bypassing_internal_transfer_limits_via_swaps
28	veridise_creditnoteproofs_can_be_stolen
29	veridise_data_escrow_encrypted_message_wrong_input
30	veridise_decoder_accepting_bogus_output_signal
31	veridise_fortxreward_abstracted_away_in_reward_calc
32	veridise_incorrect_initialization_in_membership_circuits
33	veridise_kyt_signature_verification_fails_nonzero_hash
34	veridise_missing_range_checks_in_bigmod
35	veridise_missing_range_checks_on_comparison_circuits
36	veridise_no_zero_value_validation
37	veridise_nullifier_verification_can_be_disabled
38	veridise_ownershipproofs_could_identify_issuers
39	veridise_private_information_leakage

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ID	Bug Name
40	veridise_spammers_may_slash_themselves
41	veridise_template_CoreVerifyPubkeyG1_does_not_perform_input_validation_simplified
42	veridise_underconstrained_circuit_allows_invalid_comparison
43	veridise_underconstrained_outputs_in_bitElementMulAny
44	veridise_underconstrained_outputs_in_window4
45	veridise_underconstrained_outputs_in_windowmulfix
46	veridise_underconstrained_points_in_edwards2Montgomery
47	veridise_underconstrained_points_in_montgomery2Edwards
48	veridise_underconstrained_points_in_montgomeryAdd
49	veridise_underconstrained_points_in_montgomeryDouble
50	veridise_unsafe_num2bits_254_blacklist_leaf
51	veridise_zaccount_renewal_kyc_expiry_not_validated
52	veridise_zaccount_renewal_multiple_nullifiers_same_utxo
53	veridise_zero_padding_for_sha256_in_ExpandMessageXMD_is_vulnerable_to_an_overflow
54	veridise_zone_id_inclusion_prover_bypass
55	veridise_zone_limits_bypass_zswapv1
56	yacademy_Knowledge_of_any_member_signature_allow_to_generate_proof_of_membership
57	yacademy_input_signal_s_is_not_constrained_in_eff_ecdsa_circom
58	yacademy_under_constrained_circuits_compromising_the_soundness_of_the_system
59	zksecurity_Unsound_Addition_Gadget
60	zksecurity_Unsound_XOR_gadget
61	zksecurity_an_attacker_can_craft_a_fake_non_inclusion_proof_for_a_given_key_due_to_an_aliasing_bug_in_the_smt_verifier
62	zksecurity_big_integer_zero_check_is_not_sound
63	zksecurity_exclusion_check_of_forbidden_countries_is_unsound_and_incomplete_due_to_incorrect_indexing
64	zksecurity_forbidden_country_check_bypass_via_packed_byte_overflow
65	zksecurity_missing_boolean_constraints_in_the_merkle_tree_path_leads_to_an_attacker_being_able_to_craft_a_fake_merkle_proof_for_an_arbitrary_L
66	zksecurity_missing_byte_range_checks_allows_packed_data_pollution
67	zksecurity_second_pre_image_attacks_on_packbytesandposeidon_may_be_used_to_register_arbitrary_passports_and_dsc_certificates
68	zksecurity_sha256_templates_return_zero_on_arbitrary_inputs
69	zksecurity_the_registration_and_disclosure_circuits_lack_range_checks_for_the_input_indices
70	zksecurity_unsound_left_rotation

Table 5: Execution Times per Bug (by Bug ID, in seconds)

Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
1 0.24	0.02	0.15	1.85	22.80	89.63
2 —	0.30	—	333.50	—	—
3 6.01	0.01	—	—	20.73	0.04
4 0.37	0.01	0.14	1.19	13.40	0.03
5 0.78	0.01	0.13	238.67	23.09	0.02
6 0.26	0.01	—	1.48	17.50	—
7 2.86	0.01	—	—	18.17	—
8 19.58	0.02	—	2.61	27.04	—
9 0.24	0.01	0.07	2.31	25.55	0.02
10 0.53	0.03	—	—	26.94	0.13
11 0.28	0.01	0.29	2.62	23.19	2.63
12 58.59	0.18	—	10.77	48.44	—
13 6.86	0.16	—	4.36	32.31	—
14 —	0.02	66.86	—	20.95	1.38
15 288.16	0.02	249.13	—	18.93	—
16 544.52	0.02	49.64	—	16.06	0.97
17 30.45	0.02	507.20	—	23.56	15.01
18 27.62	0.04	0.88	446.72	23.64	—
19 —	0.02	64.33	21.59	76.66	—
20	0.06	—	—	—	42.87

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
214.15					
21	0.18	—	2.07	—	2.45
0.44					
22	0.23	—	—	—	9.71
85.62					
23	0.01	500.26	—	19.82	—
0.44					
24	0.06	0.13	1.32	15.82	0.12
0.25					
25	0.10	—	3.48	29.35	6.79
0.23					
26	0.13	—	2.94	22.41	—
2.89					
27	0.09	—	3.53	28.22	461.40
0.29					
28	0.02	—	3.85	22.89	—
16.97					
29	0.26	—	17.80	91.09	—
139.63					
30	0.01	0.04	1.35	18.08	0.09
0.40					
31	0.10	—	1.87	21.03	0.10
0.35					
32	0.04	0.06	1.98	27.61	16.65
0.23					
33	0.32	—	444.92	—	39.30
—					
34	0.03	—	—	27.53	—
163.57					
35	0.01	504.09	1.16	12.93	—
4.94					
36	0.19	13.99	4.85	34.13	—
—					
37	0.29	—	439.58	—	40.47
—					
38	0.07	—	1.11	13.70	—
0.28					
39	0.11	—	7.91	44.08	—
42.71					

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
40 0.99	0.10	4.02	1.54	15.08	—
41 40.36	0.24	501.91	—	34.89	0.86
42 33.43	0.03	553.48	—	18.06	—
43 111.23	0.02	502.19	81.75	16.91	0.05
44 20.80	0.04	502.33	338.44	20.02	3.22
45 20.94	0.02	502.30	338.24	20.33	1.69
46 0.32	0.01	0.09	1.59	18.91	0.41
47 0.37	0.01	0.04	1.34	16.19	0.08
48 0.90	0.01	2.06	1.43	16.31	0.01
49 —	0.01	500.10	1.48	16.74	0.01
50 3.19	0.15	—	3.45	25.29	—
51 509.59	0.26	—	55.23	141.61	86.32
52 496.19	0.24	—	56.28	142.66	99.27
53 0.39	0.06	500.16	—	22.96	531.69
54 0.30	0.04	—	1.53	18.87	0.16
55 —	0.33	—	454.02	328.75	41.22
56 6.59	0.05	—	3.88	33.87	—
57 —	0.04	—	—	24.90	—
58 23.19	0.03	—	—	25.84	0.22
59	0.01	0.07	1.69	22.30	0.02

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
0.26 60	0.01	500.26	494.72	22.21	0.03
0.32 61	0.01	—	—	19.49	—
517.44 62	0.01	—	1.08	14.07	17.04
0.29 63	0.01	—	1.08	15.92	71.24
0.39 64	0.03	—	1.47	17.31	74.86
0.36 65	0.01	—	104.05	18.91	—
33.73 66	0.03	—	—	27.95	—
71.31 67	0.02	—	1.60	13.21	—
3.93 68	0.04	—	—	148.34	—
— 69	0.01	—	1.47	17.48	0.02
0.31 70	—	19.04	2.53	22.17	0.84
0.27					